دانشگاه زنجان

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Vision Transformers (*ViT*) and Their Modern Variants in Image Classification

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Introduction / Motivation

Vision Transformers (ViT) have recently emerged as one of the most popular and widely cited approaches in computer vision. Inspired by Transformer architectures from natural language processing, ViTs have demonstrated highly competitive performance in image classification tasks, sometimes surpassing traditional convolutional neural networks (CNNs).

This research project aims to investigate ViTs and their modern variants, analyzing their architectures, comparing their performance, and running practical implementations on small datasets in Google Colab. The motivation for selecting this topic is the combination of high academic relevance, strong citation counts, and practical feasibility for a student project.

Topic Description

Vision Transformers (ViT) apply the self-attention mechanism of Transformers to image data. Images are divided into patches, each patch is flattened and embedded, and the sequence of patch embeddings is processed by a standard Transformer encoder. The output class token is then used for classification.

Modern variants of ViT include:

- **DeiT (Data-efficient Image Transformers)** – a lighter, more efficient version of ViT
- **Swin Transformer** – hierarchical structure for better local representation
- **MobileViT** – optimized for mobile and edge devices

These variants allow flexibility in implementation and experimentation, making the topic highly suitable for research and code execution in Google Colab.

Research Goal

Vision Transformers (ViT) apply the self-attention mechanism of Transformers to image data. Images are divided into patches, each patch is flattened and embedded, and the sequence of patch embeddings is processed by a standard Transformer encoder. The output class token is then used for classification.

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Main Survey Paper

Paper: *Transformers in Vision: A Survey*
Authors: Khan, Naseer, Hayat, Zamir, Shahbaz Khan & Shah
Year: 2021
Citations: ~2,200
Link: https://arxiv.org/abs/2101.01169

Description / Justification

This paper is selected as the main survey because it provides a comprehensive overview of Vision Transformers (ViTs) and their applications in computer vision, including:

- Classification
- Detection
- Segmentation
- Video and multimodal tasks

It also discusses architecture, variants, advantages, limitations, and future directions. The paper is highly cited (~2,200 citations), making it academically credible, and is well-structured and readable, which simplifies understanding and summarization.

Supporting Papers

1. *A Survey on Efficient Vision Transformers: Algorithms, Techniques, and Performance Benchmarking (2023) https://arxiv.org/abs/2309.02031*
2. *Vision Transformers for Image Classification: A Comparative Survey (2025) https://doi.org/10.3390/technologies13010032*
3. *Exploring the Synergies of Hybrid CNNs and ViTs Architectures for Computer Vision: A survey (2024) https://arxiv.org/abs/2402.02941*